Boiler combustion efficiency analysis

Boiler efficiency may be calculated using the loss method, i.e.:

$$\eta = 100 - \sum q_i \ \ [\%],$$

where $\sum q_i = q_1 + q_2 + q_3$ are the measurable losses. Only the most significant losses are considered.

(i) **Exhaust gases heat loss** is the function of the enthalpy (temperature) of the exhaust gases. The simplified Siegert formula is used to calculate it:

$$q_1 = \sigma \cdot \frac{(t_{hg} - t_{am}) + 0.59 \cdot CO}{CO_2 + CO} \ \ [\%],$$

where:

t_{hg} – exhaust gases temperature;
t_{am} – ambient air temperature = 25°;
σ – Siegert parameter, assumed value = *1.25* ;
CO – CO amount in flue gases in [%];
CO_2 – CO_2 amount in flue gases in [%].
CO_2 can be approximated with Siegerta formula:

$$CO_2 = CO_{2max} \cdot \left(1 - \frac{O_2}{20.95}\right)$$

where:

CO_{2max} – Siegert coefficient,
O_2 – oxygen concentration in flue gases.

(ii) **Chemical CO incomplete combustion loss** is calculated using the following formula:

$$q_2 = \alpha \cdot \frac{CO}{CO_2 + CO},$$

where:

α – Siegert coefficient;
CO – CO amount in flue gases in [%];
CO_2 – CO_2 amount in flue gases in [%].

(iii) **Conduction heat loss** is specific for the boiler construction and its exact calculation is very difficult to be determined. Experience shows only strong dependences of the loss on steam power. There is approximate equation for that:

$$q_3 = \frac{D_{sh}^{nom}}{D_{sh}} \cdot \sqrt{\frac{100}{D_{sh}^{nom}}}$$

where:

D_{sh}^{nom} – nominal steam flow [T/h]; D_{sh} – steam flow [T/h].